vector3d

A small 3D vector class for the *OOP in Physics* course. Internals are covered in Week 4.

Import

```python
from vector3d import Vector3D
```

Operations

```python
a = Vector3D(1.0, 2.0, 3.0)
b = Vector3D(0.5, 0.0, -1.0)
```

Operation	Example	Result
Create	`Vector3D(1.0, 2.0, 3.0)`	`(1.0, 2.0, 3.0)`
Components	`a.x, a.y, a.z`	`1.0 2.0 3.0`
Add	`a + b`	`(1.5, 2.0, 2.0)`
Subtract	`a - b`	`(0.5, 2.0, 4.0)`
Scalar multiply	`a * 2.0` or `2.0 * a`	`(2.0, 4.0, 6.0)`
Scalar divide	`a / 2.0`	`(0.5, 1.0, 1.5)`
Negate	`-a`	`(-1.0, -2.0, -3.0)`
Norm	`abs(b)`	`1.118...`
Dot product	`a.dot(b)`	`-2.5`
Cross product	`a.cross(b)`	`(-2.0, 2.5, -1.0)`
Print	`print(a)`	`(1.0, 2.0, 3.0)`
Repr	`a`	`Vector3D(1.0, 2.0, 3.0)`

Not available

- `a == b` compares identity, not components; it is `False` even for equal vectors.
- `s / a` and `s + a` raise `TypeError`.
- `a * b` (vector times vector) does **not** raise; it silently returns a tuple of vectors, which is meaningless. Use `a.dot(b)` or `a.cross(b)`.

Physics example

python

```python
r = Vector3D(1.0, 2.0, 3.0)   # position, m
v = Vector3D(0.5, 0.0, -1.0)  # velocity, m/s

r_new = r + v * 0.1           # position after 0.1 s
L = r.cross(v) * 2.0          # angular momentum, m = 2 kg
```